

4 Interpretations of Probability and Statistics

- There are two main interpretations of probability: relative frequency and “subjective” probability.
- These two interpretations provide the philosophical foundation for two schools of statistics: frequentist and Bayesian

Example 4.1. The following situations involve probability. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of “probability” the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

1. The probability that you roll doubles in a turn of a board game.
2. The probability that a “randomly selected” Cal Poly student is a California resident.
3. The probability that the high temperature in San Luis Obispo tomorrow is above 70 degrees F.
4. The probability that a Democratic candidate wins the 2028 U.S. Presidential election.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2009.

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur. Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event’s likelihood or degree of uncertainty.
- The probability of an event associated with a random phenomenon can be interpreted as a **long run proportion** or **long run relative frequency**: the probability of the event is the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon, e.g., in a simulation.
- A **subjective (a.k.a. personal) probability** describes the degree of likelihood a given individual assigns to a certain event. As the name suggests, different individuals (or probabilistic models) might have different subjective probabilities for the same event.

- **Think of subjective probabilities as measuring *relative degrees of likelihood or uncertainty*** rather than long run relative frequencies.
- We will use these interpretations interchangeably.
 - With subjective probabilities it is often helpful to consider what might happen in a simulation.
 - It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.

Example 4.2. What is your subjective probability that Professor Ross has ever attended a “famous” sporting event (like the Superbowl, a World Series game 7, an NCAA tournament game with a famous shot, an Olympic final, etc)? Consider the following two bets, and suppose you must choose only one.

- A. You win \$100 if Professor Ross has attended a famous sporting event, and you win nothing otherwise.
 - B. A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win \$100 if the selected marble is green, and you win nothing otherwise.
1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?
 2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?
 3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 4.3. Suppose your subjective probabilities for the 2026 NBA champion satisfy the following conditions.

- The Nuggets and Spurs are equally likely to win
- The Celtics are 1.5 times more likely than the Nuggets to win
- The Thunder are 2 times more likely than the Celtics to win
- The winner is as likely to be among these four teams — Thunder, Celtics, Spurs, Nuggets — as not.

1. Construct a table of your subjective probabilities.
2. How could you construct a circular spinner (like from a kids game) to simulate the NBA champion according to these subjective probabilities? According to this model, what would you expect the results of 10000 repetitions of a simulation of the champion to look like?

Example 4.4. How many miles are on the odometer of Professor Ross's car? Consider mileage on a continuous scale.

In this example, you will use probability to quantify your uncertainty about the mileage. You only need to give ballpark estimates of your subjective probabilities, but you might consider what kinds of bets you would be willing to accept.

1. What is your subjective probability that the mileage is at most 50000? More than 50000? (What must be true about these two probabilities?) You're welcome to try with values other than 50000.

2. Fill in the blank: your subjective probability that the mileage is at most [blank] years old is equal to 50%.
 3. Fill in the blanks: your subjective probability that the mileage is between [blank] and [blank] years old is equal to 98%.
 4. Let θ represent the current mileage. Use your answers to the previous parts to sketch a continuous probability density function to represent your *subjective probability distribution* for θ .
 5. If you ascribe a probability distribution to θ , then are you treating θ as a constant or a random variable?
- A **random variable** is a numerical quantity whose value is determined by the outcome of a random or uncertain phenomenon.
 - The **(probability) distribution** of a random variable specifies the possible values of the random variable and a way of determining corresponding probabilities. Like probabilities themselves, probability distributions of random variables can also be interpreted as:
 - *relative frequency distributions*, e.g., what pattern would emerge if I simulated many values of the random variable? or as
 - *subjective probability distributions*, e.g., which potential values of this uncertain quantity are relatively more plausible than others?

1. Since mileage is treated as a continuous random variable, each of the above plots is a probability “density”. Explain briefly what this means. How is probability represented in density plots like these?

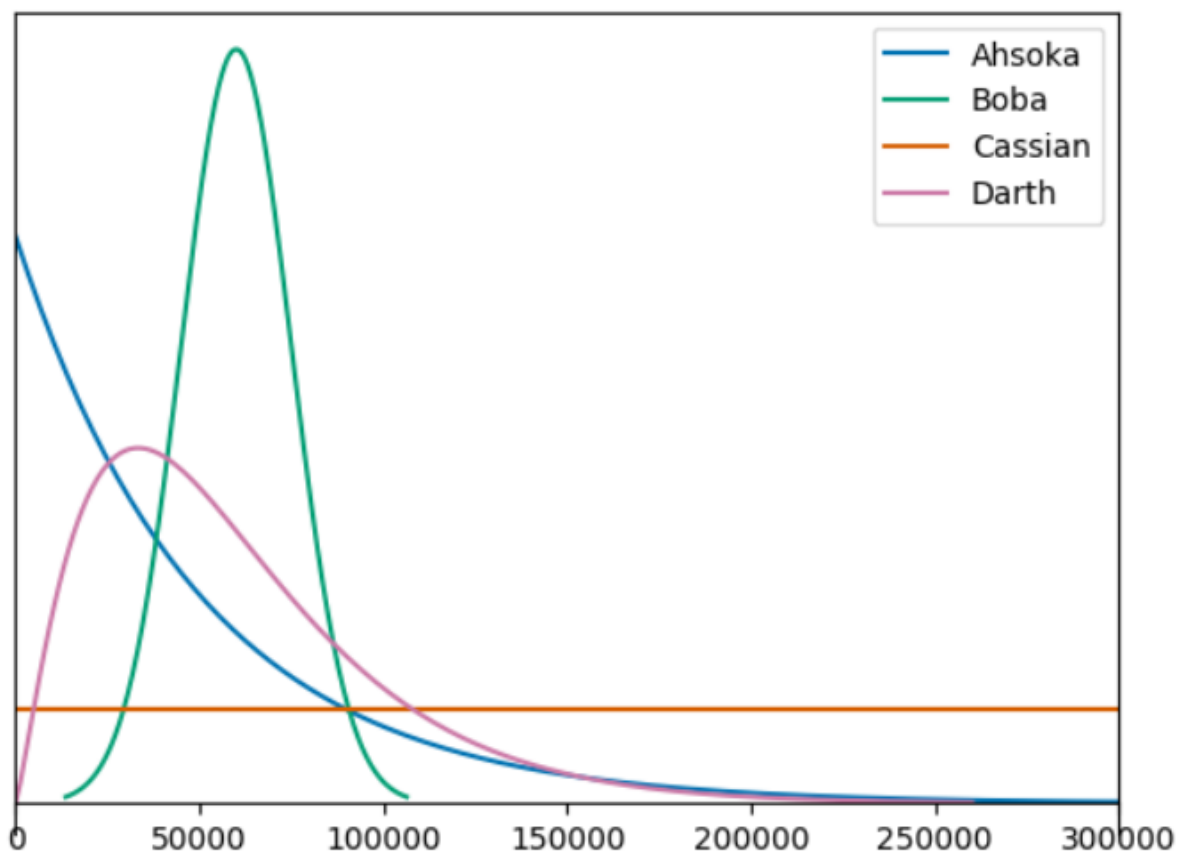


Figure 4.1: Plausibility assessments of four students

2. Rank the students in terms of their subjective probability that the mileage is at most 50000.

3. Rank the students in terms of their uncertainty about the mileage. Who is the most uncertain? The least?

- One key aspect of Bayesian analyses is *applying a subjective probability distribution to a parameter* in a statistical model.

Example 4.6. Let θ_b represent the proportion of current Cal Poly students who have ever read at least one book in Lemony Snicket's *A Series of Unfortunate Events*. Let θ_m represent the proportion of current Cal Poly students who have seen at least one Star Wars movie.

1. Are θ_b and θ_m parameters or statistics? Why?

2. Are the values of θ_b and θ_m known or unknown, certain or uncertain?

3. What are the possible values of θ_b and θ_m ?

4. Sketch a probability distribution representing what you think are more/less credible values of θ_b . Repeat for θ_m .

 5. Are you more certain about the value of θ_b or θ_m ? How is this reflected in your distributions?

 6. Suppose that in a class of 35 Cal Poly students, 7 have read a Lemony Snicket book, and 32 have seen a Star Wars movie. Now that we have observed some data, sketch a probability distribution representing what you think are more/less credible values of θ_b . Repeat for θ_m . How do your distributions after observing data compare to the distributions you sketched before?
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- **Observational units** (a.k.a., cases, individuals, subjects) are the people, places, things, etc we collect information on.
 - A **variable** is any characteristic of an observational unit that we can measure.
 - **Statistical inference** involves using data collected on a *sample* to make conclusions about a *population*.
 - Inference often concerns specific numerical summaries, using values of *statistics* to make conclusions about *parameters*.
 - A **statistic** is a number that describes the **sample**, e.g., *sample mean*, *sample proportion*.
 - A **parameter** is a number that describes the **population** or model, e.g., *population mean*, *population proportion*. The actual value of a parameter is almost always *unknown*.
 - Parameters are often denoted with Greek letters. We'll often use the Greek letter θ (“theta”) to denote a generic parameter.
 - In “traditional”, *frequentist* statistical analysis, parameters are treated as *fixed* — *that is, not random* — *constants*.
 - Any randomness in a frequentist analysis arises from how the data were collected, e.g., via random sampling or random assignment.

- In a frequentist analysis, statistics are random variables; parameters are fixed numbers.
- On the other hand, in a Bayesian statistical analysis, since a parameter θ is unknown — that is, its value is *uncertain* to the observer — θ is treated as a *random variable*.
 - That is, **in Bayesian statistical analyses unknown parameters are random variables that have probability distributions.**
 - The (subjective) probability distribution of a parameter quantifies the degree of uncertainty about the value of the parameter.
 - Therefore, the Bayesian perspective allows for probability statements about parameters.
 - We start with “prior” distributions that represent our uncertainty based on our “beliefs”, then we revise to obtain “posterior” distributions after observing some data.
 - We will see (among other things) (1) how to quantify uncertainty about parameters using probability distributions, and (2) how to update those distributions to reflect new data.